We use the `is_prime` function to make a table of the first few Mersenne primes (see Section 1.2.3).

```
sage: for p in primes(100):
...   if is_prime(2^p - 1):
...       print p, 2^p - 1
2 3
3 7
5 31
7 127
13 8191
17 131071
19 524287
31 2147483647
61 2305843009213693951
89 618970019642690137449562111
```

There is a specialized test for primality of Mersenne numbers called the Lucas-Lehmer test. This remarkably simple algorithm determines provably correctly whether or not a number $2^p - 1$ is prime. We implement it in a few lines of code and use the Lucas-Lehmer test to check for primality of two Mersenne numbers:

```
sage: def is_prime_lucas_lehmer(p):
...    s = Mod(4, 2^p - 1)
...    for i in range(3, p+1):
...        s = s^2 - 2
...    return s == 0
sage: # Check primality of 2^9941 - 1
sage: is_prime_lucas_lehmer(9941)
True
sage: # Check primality of 2^next_prime(1000)-1
sage: is_prime_lucas_lehmer(next_prime(1000))
False
```

For more on Mersenne primes, see the Great Internet Mersenne Prime Search (GIMPS) project at `http://www.mersenne.org/`.

2.5 The Structure of $(\mathbf{Z}/p\mathbf{Z})^*$

This section is about the structure of the group $(\mathbf{Z}/p\mathbf{Z})^*$ of units modulo a prime number p. The main result is that this group is always cyclic. We will use this result later in Chapter 4 in our proof of quadratic reciprocity.

Definition 2.5.1 (Primitive Root)**.** A *primitive root* modulo an integer n is an element of $(\mathbf{Z}/n\mathbf{Z})^*$ of order $\varphi(n)$.

We will prove that there is a primitive root modulo every prime p. Since the unit group $(\mathbf{Z}/p\mathbf{Z})^*$ has order $p-1$, this implies that $(\mathbf{Z}/p\mathbf{Z})^*$ is a cyclic group, a fact that will be extremely useful, since it completely determines the structure of $(\mathbf{Z}/p\mathbf{Z})^*$ as a group.

If n is an odd prime power, then there is a primitive root modulo n (see Exercise 2.28), but there is no primitive root modulo the prime power 2^3, and hence none mod 2^n for $n \geq 3$ (see Exercise 2.27).

Section 2.5.1 is the key input to our proof that $(\mathbf{Z}/p\mathbf{Z})^*$ is cyclic; here we show that for every divisor d of $p-1$ there are exactly d elements of $(\mathbf{Z}/p\mathbf{Z})^*$ whose order divides d. We then use this result in Section 2.5.2 to produce an element of $(\mathbf{Z}/p\mathbf{Z})^*$ of order q^r when q^r is a prime power that exactly divides $p-1$ (i.e., q^r divides $p-1$, but q^{r+1} does not divide $p-1$), and multiply together these elements to obtain an element of $(\mathbf{Z}/p\mathbf{Z})^*$ of order $p-1$.

SAGE Example 2.5.2. Use the `primitive_root` command to compute the smallest positive integer that is a primitive root modulo n. For example, below we compute primitive roots modulo p for each prime $p < 20$.

```
sage: for p in primes(20):
...     print p, primitive_root(p)
2 1
3 2
5 2
7 3
11 2
13 2
17 3
19 2
```

2.5.1 Polynomials over $\mathbf{Z}/p\mathbf{Z}$

The polynomials $x^2 - 1$ has four roots in $\mathbf{Z}/8\mathbf{Z}$, namely 1, 3, 5, and 7. In contrast, the following proposition shows that a polynomial of degree d over a field, such as $\mathbf{Z}/p\mathbf{Z}$, can have at most d roots.

Proposition 2.5.3 (Root Bound). *Let $f \in k[x]$ be a nonzero polynomial over a field k. Then there are at most $\deg(f)$ elements $\alpha \in k$ such that $f(\alpha) = 0$.*

Proof. We prove the proposition by induction on $\deg(f)$. The cases in which $\deg(f) \leq 1$ are clear. Write $f = a_n x^n + \cdots a_1 x + a_0$. If $f(\alpha) = 0$, then

$$\begin{aligned} f(x) &= f(x) - f(\alpha) \\ &= a_n(x^n - \alpha^n) + \cdots + a_1(x - \alpha) + a_0(1 - 1) \\ &= (x - \alpha)(a_n(x^{n-1} + \cdots + \alpha^{n-1}) + \cdots + a_2(x + \alpha) + a_1) \\ &= (x - \alpha)g(x), \end{aligned}$$

for some polynomial $g(x) \in k[x]$. Next, suppose that $f(\beta) = 0$ with $\beta \neq \alpha$. Then $(\beta - \alpha)g(\beta) = 0$, so, since $\beta - \alpha \neq 0$ and k is a field, we have $g(\beta) = 0$. By our inductive hypothesis, g has at most $n-1$ roots, so there are at most $n-1$ possibilities for β. It follows that f has at most n roots. $\square$

SAGE Example 2.5.4. We use Sage to find the roots of a polynomials over $\mathbf{Z}/13\mathbf{Z}$.

```
sage: R.<x> = PolynomialRing(Integers(13))
sage: f = x^15 + 1
sage: f.roots()
[(12, 1), (10, 1), (4, 1)]
sage: f(12)
0
```

The output of the roots command above lists each root along with its multiplicity (which is 1 in each case above).

Proposition 2.5.5. *Let p be a prime number and let d be a divisor of $p-1$. Then $f = x^d - 1 \in (\mathbf{Z}/p\mathbf{Z})[x]$ has exactly d roots in $\mathbf{Z}/p\mathbf{Z}$.*

Proof. Let $e = (p-1)/d$. We have

$$\begin{aligned} x^{p-1} - 1 &= (x^d)^e - 1 \\ &= (x^d - 1)((x^d)^{e-1} + (x^d)^{e-2} + \cdots + 1) \\ &= (x^d - 1)g(x), \end{aligned}$$

where $g \in (\mathbf{Z}/p\mathbf{Z})[x]$ and $\deg(g) = de - d = p - 1 - d$. Theorem 2.1.20 implies that $x^{p-1} - 1$ has exactly $p-1$ roots in $\mathbf{Z}/p\mathbf{Z}$, since every nonzero element of $\mathbf{Z}/p\mathbf{Z}$ is a root! By Proposition 2.5.3, g has *at most* $p - 1 - d$ roots and $x^d - 1$ has at most d roots. Since a root of $(x^d - 1)g(x)$ is a root of either $x^d - 1$ or $g(x)$ and $x^{p-1} - 1$ has $p - 1$ roots, g must have exactly $p - 1 - d$ roots and $x^d - 1$ must have exactly d roots, as claimed. $\square$

SAGE Example 2.5.6. We use Sage to illustrate the proposition.

```
sage: R.<x> = PolynomialRing(Integers(13))
sage: f = x^6 + 1
sage: f.roots()
[(11, 1), (8, 1), (7, 1), (6, 1), (5, 1), (2, 1)]
```

We pause to reemphasize that the analog of Proposition 2.5.5 is false when p is replaced by a composite integer n, since a root mod n of a product of two polynomials need not be a root of either factor. For example, $f = x^2 - 1 = (x-1)(x+1) \in \mathbf{Z}/15\mathbf{Z}[x]$ has the four roots 1, 4, 11, and 14.

2.5.2 Existence of Primitive Roots

Recall from Section 2.1.2 that the *order* of an element x in a finite group is the smallest $m \geq 1$ such that $x^m = 1$. In this section, we prove that $(\mathbf{Z}/p\mathbf{Z})^*$ is cyclic by using the results of Section 2.5.1 to produce an element of $(\mathbf{Z}/p\mathbf{Z})^*$ of order d for each prime power divisor d of $p-1$, and then we multiply these together to obtain an element of order $p-1$.

We will use the following lemma to assemble elements of each order dividing $p-1$ to produce an element of order $p-1$.

Lemma 2.5.7. *Suppose $a, b \in (\mathbf{Z}/n\mathbf{Z})^*$ have orders r and s, respectively, and that $\gcd(r,s) = 1$. Then ab has order rs.*

Proof. This is a general fact about commuting elements of any group; our proof only uses that $ab = ba$ and nothing special about $(\mathbf{Z}/n\mathbf{Z})^*$. Since

$$(ab)^{rs} = a^{rs}b^{rs} = 1,$$

the order of ab is a divisor of rs. Write this divisor as r_1s_1 where $r_1 \mid r$ and $s_1 \mid s$. Raise both sides of the equation

$$a^{r_1s_1}b^{r_1s_1} = (ab)^{r_1s_1} = 1$$

to the power $r_2 = r/r_1$ to obtain

$$a^{r_1r_2s_1}b^{r_1r_2s_1} = 1.$$

Since $a^{r_1r_2s_1} = (a^{r_1r_2})^{s_1} = 1$, we have

$$b^{r_1r_2s_1} = 1,$$

so $s \mid r_1r_2s_1$. Since $\gcd(s, r_1r_2) = \gcd(s,r) = 1$, it follows that $s = s_1$. Similarly $r = r_1$, so the order of ab is rs. $\square$

Theorem 2.5.8 (Primitive Roots)**.** *There is a primitive root modulo any prime p. In particular, the group $(\mathbf{Z}/p\mathbf{Z})^*$ is cyclic.*

Proof. The theorem is true if $p = 2$, since 1 is a primitive root, so we may assume $p > 2$. Write $p-1$ as a product of distinct prime powers $q_i^{n_i}$:

$$p - 1 = q_1^{n_1}q_2^{n_2}\cdots q_r^{n_r}.$$

By Proposition 2.5.5, the polynomial $x^{q_i^{n_i}} - 1$ has exactly $q_i^{n_i}$ roots, and the polynomial $x^{q_i^{n_i-1}} - 1$ has exactly $q_i^{n_i-1}$ roots. There are $q_i^{n_i} - q_i^{n_i-1} =$